

Questions Collection - Web Technology (CSIT)

1- Introduction (3hr)- PYQs

5 marks

- explain the architectural issues of web layer with example.
- explain the 2-tier, 3-tier and n-tier technology with its architecture.
- how does a web server link physically on the internet? how do we navigate from one URL to another from a page displayed at a browser? explain.
- explain the web server concept. why internet is a client/server technology?
- define HTTP. explain the overall function of HTTP with example.
- differentiate between 2-tier, 3-tier and n-tier.
- explain the client tier with example.
- list the protocols and their use at the application layer in the internet. why world wide web is used? explain.
- what is FTP? discuss its functions in detail.
- differentiate between HTTP and FTP. discuss 3-tier technology with suitable example.
- what is HTTP? discuss its function in detail.
- what is client/server technology? discuss internet as a client/server technology.
- what is client/server technology? differentiate between web client and web server.
- list the common application of web server.
- describe client server architecture with its types. *(also asked explicitly in 2081)*
- explain the uses of a web server.
- explain SMTP with example.
- write short notes on various services offered by the internet.
- explain the functions of web caches.
- what is the functionality and purpose of HTTP?
- what are the different types of headers in an HTTP message from a client? explain.
- what is HTTP protocol? define HTTP Request and Response. [2+3]
- describe HTTP with its header fields.
- explain the client/server concepts of web.
- what is FTP?
- what is HTTP?
- explain about domain name and domain name system.
- discuss different types of client server architectures.
- what is web server software? discuss its role on the internet.
- what is web client? discuss its role on the internet.
- what are the essential features of web browser? explain.

- explain briefly, how the domain names are translated to IP addresses.
- differentiate between domain name and domain name system.
- explain the client/server concepts of web.
- describe the significance of IP addresses and their types.
- write short notes on: a) SMTP, b) XQuery
- difference between static and dynamic website.
- describe the www, web client and web server. (asked 2080)
- what are the services provided under web2.0? (asked 2078)

2 - HTML (10)- PYQs

10 marks

1. explain the structure of html file with example. how is html different from css? explain three different ways for inserting css code into an html file.
2. define dtd. what are its applications? create the html document with the paragraph using `<p>`, `<h1>`, `<strong>` for the first word for every sentence and `<em>` for all the capital letters.
3. using test structuring toys, create a homepage for the curriculum vitae. save at web server that is local or on the internet. later view it using the browser.
4. create html signup form with fields name, email, password, age. validate using js: all fields not empty, email valid, password at least 6 chars starting with alphabet ending with digit, age between 8-60. (asked on 2078)
5. Write HTML script to create HTML form as below and perform form validation using JavaScript. The validation should be as; the username should be of length at least 4 and is required. The password should start with a character and end with # and is required. The country and gender should be checked. (asked on 2080)

5 marks

- what are the tags and attributes for a table in html document?
- what is the structure of html file? explain with example.
- what is html? differentiate between "get" and "post" methods used to send form data.
- discuss the structure of html document with suitable example.
- how can you draw a table in html? discuss with suitable example.
- write a html script to insert audio file in a html page? use the controls and autoplay properties.
- write an html code to display a simple form for user login.
- what are the color attributes in html? explain.
- write an html code to display the following:

ID	Name	Address	Age

- write html script for displaying table with following output.

Drinks	Snacks
1. Coke	Momo
2. Sprite	
• Milk	Chowmin
• Coffee	

- write an html code to display the following:

Example
$(a + b)^2 = a^2 + 2ab + b^2$
Thank You

- write an html code to display the following:

Username:

Password:

- what are the attributes of list tag? explain.
- what are the visibility and display properties of html elements? explain with example.
- create a html page with tags header, article and footer. insert a link containing <mailto:info@iost.edu.np> in the footer tag. set the keywords "iost", "csit" using meta tag in the page.
- write html script for displaying div with following output. the div should be floating right.

CSIT
1. PMC
2. BMC
BCA
• TC
• PPL

- why inline frames are used? create a html page containing iframe within a paragraph. the iframe has source to <http://www.tuiost.edu.np> and its height and width are 200px and 400px.
- write a html script to create a table with 3 rows and 3 columns. the second row third column should contain text "tu" with link to www.tu.edu.np. set height of second row to 200px. *(asked on 2081)*
- create a html page with link to <mailto:info@iost.edu.np> in footer. set keywords "iost", "csit" with meta tag.
- create registration form with name, country, gender, submit. display "registered" on keypress.
- write html script for form with username, password, checkbox. validate with javascript: username not empty, password at least 5 chars, checkbox checked.
- create html page with a div (id="dv1") containing canvas (id="mycanvas", height=200, width=300). *(asked on 2079)*
- write html to generate list:
 1. coffee : I. black coffee II. green coffee
 2. tea A: black tea B. green tea*(asked on 2079)*
- why inline frames are used? create html with iframe inside paragraph with src=<http://www.tuiost.edu.np>, height=200px, width=400px. *(asked on 2076)*
- create html with header, article, footer, mailto link to info@iost.edu.np, meta keywords "iost", "csit". *(asked on 2076)*
- what is http protocol? define http request and response. *(asked on 2079)*
- create html with ordered and unordered lists. set ordered list type A, start at D. *(asked on 2078)*
- create html page with body containing div with image linked to <http://www.tuiost.edu.np>. set page title "iost", add meta tag. *(asked on 2078)*

3- CSS (8 hr)- PYQs

10 marks

- How positioning of HTML elements can be done using CSS? Create a HTML page with a div with some content and two paragraph with some contents having id p1 and p2. Write external CSS for the div tag having fixed position with border style solid and width 2 px. the p1 should have relative position. The font type of p1 should be Arial and color should be red. The p2 have absolute position with top 0px and left 200px. *(asked on 2076 exam)*
- How external CSS is inserted in an HTML page? Create an HTML page with two paragraph tags With id p1 and p2, and one div tag with id dv1. Write an external CSS file as per the description. the margin of the paragraphs should be 100px from the top and 80px from the left. Both of paragraphs should have the display set to block. The dv1 should have padding left and right set to 30px and 25px respectively, its background color should be blue and the font color should be red with the font type Verdana. Set the display of dv1 to none. *(asked on 2078 exam)*

5 marks

- How internal and external CSS differ from each other. Illustrate with example. *(asked on 2081 exam)*
- What is text-overflow property in CSS ? How can you use clip and ellipsis in text-overflow ? *(asked on 2081 exam)*
- Explain CSS box model. *(asked on 2080 exam)*

- How can you define position property for specifying specific positioning of an HTML element? Write internal CSS to illustrate position of an HTML div. *(asked on 2080 exam)*
- What are the usages of class and id selectors in CSS? Illustrate with example. *(asked on 2079 exam)*
- Describe CSS Box model with example. *(asked on 2078 exam)*
- Explain with example the basic concept of cascading and inheritance in CSS.
- How external style sheets can be inserted in a HTML document? Illustrate with example.
- Define CSS? How can you use an external CSS? Explain.
- What is CSS? How can you use an internal CSS? Explain.
- What is CSS selector? Compare id selector with class selector.
- How can you insert inline style sheet in the HTML document? Discuss with example.
- How can you insert internal style sheet in your HTML document? Discuss with example.
- How can you insert external style sheets in your HTML document? Discuss with example.
- Write a HTML script containing use of media query for changing the background-color of html page to black if the viewport is 600 pixels wide or more than that otherwise if the viewport is less than 500 pixels, the background-color should be changed to red.
- How XSL is different from CSS? Explain.
- Discuss about normal flow box layout in CSS and also discuss about positioning in CSS.
- How web pages can be made responsive using media queries ? Illustrate with an example. [2+3]

4- JS (9hr)- PYQs

5 marks

- What is onload event in HTML? Write an HTML script that displays alert message "Hello Nepal" in onload event. *(asked on 2081 exam)*
- How can you define array in javascript ? Create an array Fruit with its elements "apple", "banana", "mango", "grape" and print the array. *(asked on 2081 exam)*
- Write a PHP function add that takes two arguments as an input and return their sum. *(asked on 2081 exam)*
(this is PHP, seems misplaced)
- Create an HTML file containing form with textbox, select and radio button. Now write a javascript function for validating the form elements. Validate textbox for required, select field should have some value selected and radio button should be checked. *(asked on 2081 exam)*
- What is JQuery animate? Explain with an example.
- Write a client side script to display the array ["Cherry", "Mango", "Watermelon", "Strawberry"] in the body of HTML.
- What is chaining in jQuery? Write a script to illustrate concept of chaining using slideUp and slideDown method. *(asked on 2080 exam)*
- How HTML elements are accessed using getElementById and getElementsByTagName, illustrate using HTML script. *(asked on 2080 exam)*
- What are HTML keyboard events? Write a script to display a message "Hello world" on keypress event. *(asked on 2080 exam)*
- Write HTML script to create HTML form as below and perform form validation using JavaScript. The validation should be as; the username should be of length at least 4 and is required. The password should

start with a character and end with # and is required. The country and gender should be checked. (asked on 2080 exam)

Username

Password

Country

Gender ☐ M ☐ F

- Describe the jQuery Animate() method. Write an html script to create a div with id "mydiv". The position of the div should be set to absolute. Now show the use of animate method to the div element to right by 300px. (asked on 2079 exam)
- What is word-wrap property in CSS? Write an HTML script to illustrate the word-wrap property. (asked on 2079 exam)
- Differentiate Web 1.0 from Web 2.0. (asked on 2079 exam)
- Create an HTML file containing form elements textbox for username, password field and a checkbox for hobbies. Now write javascript function for the form validation. You should validate for username to be empty, password should be of length at least 5, and the check box should be checked. (asked on 2079 exam)
- How jQuery animate can be used to create custom animation? Write syntax with sample script. (asked on 2076 exam)
- How can you create objects in Javascript? Create a HTML page containing a division with image and its description in paragraph. Write a javascript function to change the value of description in the paragraph during onmouseover and mouseout event on the image. *(asked on 2076 exam)***
- Discuss about different JSON data types. (asked on 2078 exam)
- How jQuery id selector can be used to select specific element? Write an example with jQuery that will hide a paragraph on clicking a button. (asked on 2078 exam)
- What do you mean by cookie? Write a Javascript code to handle cookie?
- What are the two traditional ways of assigning event handlers in DOM? Explain.
- How can you define event handler? Write a JavaScript to demonstrate event handling.
- Explain about to charge an HTML element with the help of DOM.
- What are the important uses of cookies? Give some examples.
- Write a JavaScript code to create an array of elements {csit, it, bca, bim}. Display the array in body of HTML.
- Write a simple client side script to set and retrieve cookies.
- Write an example of client side script to validate a form.
- Discuss the use of Cookies with suitable example.
- Discuss different types of jQuery Selectors with examples.
- Explain about client side programming with example?
- What is the use of JSON? How can you parse a JSON, illustrate with an example. [1+4]
- What do you mean by cookies? Explain with example.
- Write short notes on:
 - a. HTML DOM
 - b. Tag libraries

10 marks

- Write HTML Script for creating a form containing text box for username, password field for password, and checkbox for Education fields. Write a JavaScript function for the validation of the form for all of the fields as required. In addition length of username should at least 4, the password should start with digit and end with \$.
- Design a form containing text box for username, password field for password, radio button for gender and checkbox for hobbies. Write a program for client-side validation of the form for the username and password field as required fields, length of username should be 5, the radio button and checkbox should be checked.
- What is DOM Hierarchy? Explain the use of , ? and + in defining a DOM element.
- Design DOM for the time table of your classes in the present semester. Save at web server that is local or on the Internet. Later view it using the browser.
- How can you create objects in Javascript? Create a HTML page containing a division with image and its description in paragraph. Write a javascript function to change the value of description in the paragraph during onmouseover and mouseout event on the image. [4+6]

5- XML (7hr)- PYQs

5 marks

- Compare XML Schema with DTD.
- What is XMLHttpRequest object? Write HTML script with AJAX showing the use of XMLHttpRequest object. (asked on 2080 exam)
- What is XML? Why is it called extensible? Mention the use of XML. (asked on 2080 exam)
- What is XML Namespace? How it is used to avoid element name conflict in XML? Justify with an example. (asked on 2079 exam)
- What is the use of XML? Create a XML file with simple type and complex type elements. Write its equivalent DTD. (asked on 2079 exam)
- How XSD of a XML file is created? Illustrate with an example. *(asked on 2076 exam)
- What is XML? Create an XML file containing records of employee having elements of simple and complex types. *(asked on 2076 exam)
- What is DTD? Create a XML file and write its equivalent DTD. *(asked on 2078 exam)
- Compare XML schema with DTD.
- What are the advantages of using XML? Explain the rules for writing XML.
- How to transform XML documents to other forms? Explain.
- Explain the XML syntax and structure rules.
- Mention the application of XSL.
- How to represent data types in XML schema? Explain.
- What are the advantages of XML over the HTML?
- What are the benefits and drawbacks of using XML namespace?
- Discuss the rules of writing XML document.
- What is XML namespace? How it is used in XML files?
- Why do we need XSLT? Explain XSL xsl:for-each Element.
- Why do we need XSLT? Explain XSLxsl:if Element.

- What are the various contents of an element in a DTD? Explain.
- Consider a XML file containing element name city <city> </city>. Now write its equivalent XSD to limit the content of city to a set of acceptable values from "PKR", "KTM", "NPJ".
- Why is it important to specify a DTD for an XML document? Explain.
- Discuss the private external declaration of DTD with example.
- What is the syntax of declaring an attributes in a DTD?
- Discuss public external declaration of DTD with example.
- How attributes and elements are defined in a DTD? Illustrate with an example.
- What are the components of an XML file? Explain.
- Write short notes on:
 - a) XML Namespace
 - b) HTML DOM
- Write short notes on:
 - a) POP
 - b) XPath

10 marks

- Describe the rules for creating XML document. Create a XML file with simple type and elements with attribute and nested elements. Write its equivalent XSD. *(asked on 2081 exam)*
- Write the structure of XML file with example. Write XML and DTD to describe "weather report" as an element and "date, location, city, state and temperature range" as its attribute.
- How XML is different from HTML? Write XML script to create XML document containing element Person having sub elements Age, Name and Pincode. Write equivalent XSD for validating restrictions on the elements as the Age should be between 10 and 20, the Name should be only accepting either Ram or Shyam and the pincode should be 3 digit. *(asked on 2080 exam)*
- Define XML schema. Compare it with DTD. Write a schema to describe "name" as a root element and "first_name", "middle_name" and "last_name" as child elements.
- What is XML? Discuss the structure of XML file with example. Write an XML schema to describe "address" as an element and "city, state, street and house-no" as its attributes.
- Discuss the benefits of using XML. Differentiate XML schema with DTD. Write an XML DTD to describe "person" as an element and "Name, address, phone-no, and Age" as its attributes.
- What are elements and attributes of XML document? What is XML schema? Differentiate XML schema with DTD. Write a simple example of DTD.
- Explain the use of XML in web development. Write internal and external DTD to describe "mail" as a root element and "to", "from", and "subject", and "message" as child elements.
- What is the use of XSLT? Create a XML file with elements and elements having attributes. Write the equivalent XSLT of the XML for rendering as a HTML document in a browser.
- Explain about XML and how to create HTML document with the help of XML.
- How can you transform XML document into HTML using XSLT. Support your answer with an example. *(asked on 2081 exam)*
- Create a small XML file designed to contain information about student performance on a module. Each student has a name, roll no, a subject mark and an exam mark.

10- PHP (8hr)- PYQs

5 marks

- Define cookie. Write a PHP program to create and retrieve a cookie. *(asked in 2081 exam)*
- How can you handle session in PHP? *(asked in 2080 exam)*
- What is the use of JSON? How can you parse a JSON, illustrate with an example. *(asked in 2076 exam)*
- How database connection is created in PHP? Write a PHP program to create a form and insert values into the database table. The form should contain at least two input fields. *(asked in 2079 exam)*
- How can you define variables in PHP? Define any two variable of string types and display their results after concatenation. *(asked in 2076 exam)*
- How can you define function in PHP? Create a function that will take two integers as argument and will return product of them. *(asked in 2078 exam)*
- How can you define array in PHP? Write a PHP function to create an array of type integer and print its elements. *(asked in 2079 exam)*
- How do you define an Array in PHP? Write a PHP script to create a multidimensional array name Product that will contain pcode, pname, and price. Initialize the array with at least three instances. Also, Write an HTML script to display the initialized values of the array in an HTML Table. *(asked in 2078 exam)*
- How can you define class and objects in PHP? Write a PHP script to create class and its object. *(asked in 2079 exam)*
- What is the importance of session?
- Why do we need session? Explain with suitable example.
- How session handling is done in server side scripts?
- Discuss anti-overload techniques in web server.
- Discuss authentication by IP addressing and domain.
- Discuss integrated windows authentication.
- Explain form handling with example.
- Differentiate between file handling and form handling.
- Write short notes on:
 - a. Tag libraries
 - b. Anonymous access
 - a. X Query *(note: asked separately as well)*
 - b. Tag libraries *(duplicate but marked)*
- How you can create dynamic content? Explain.
- Write a PHP program to create and retrieve a cookie. *(asked in 2081 exam)*
- Write a PHP program that includes a function that will take two strings as input argument and print concatenation of two strings. *(asked in 2080 exam)*
- Write a server side script in PHP to illustrate inserting and selecting values to and from database table. Create required connection using your own assumptions. *(asked in 2080 exam)*
- Write a PHP function to create an array of type integer and print its elements. *(asked in 2079 exam)*
- Write a PHP script to create class and its object. *(asked in 2079 exam)*
- Write a PHP program to create a form and insert values into the database table. The form should contain at least two input fields. *(asked in 2079 exam)*
- Write a PHP script that illustrates handling of cookie.

- Write a server-side code to select data stored in the database.
- Write a server-side code to insert data in the database.
- Write an example of server-side script to update data in the database.

10 marks

- Write a PHP program to create a form. Create an insertdb() method to insert the form values into a database table and display() to display the database table value. *(asked in 2081 exam)*
- Create web form for book search catalog. The form should contain a dropdown defining search type, a text box for search keyword, a radio button for download type true or false, now write PHP script to store data from the form into database table and also retrieve the results from stored table in a new page. *(asked in 2076 exam)*
- Write a PHP script to define two string variables, concatenate and display the result. *(asked in 2076 exam)*
- Write a PHP script to create a multidimensional array name Product that will contain pcode, pname, and price. Initialize the array with at least three instances. Also, Write an HTML script to display the initialized values of the array in an HTML Table. *(asked in 2078 exam)*
- Develop a simple website that asks the users to register using username, password, and email address. Store these information's in a database. Use client side script to validate the user input during registration.
- Develop a simple website that stores its contents in a database and displays those contents from the database. Assume at least five different contents.
- Develop a website that checks validity of a user during the log-in process. Assume that the login data (username and password) are already stored in a database. Use client-side script to validate the user input during login process.
- Construct a web page that takes name, address, and phone number from the user and stores this information in the database using server side script. Use JavaScript to validate form data for phone number.
- Develop a simple Web site that takes first name, last name and address from the user and stores this information in the database. Use JavaScript to validate form data.
- Prepare a form that should contain text box for name and email, selection list for country. Write a server side script to store data from the form into database. The script should contain connection to database and appropriate query into the database.
- Prepare a form that should contain text box, selection list and radio button. Write PHP script to store data from the form into database using database connection and appropriate query.

reference:

[collegenote](#)

[hamrocsit](#)